

- 2 (a) State what is meant by a *field of force*.

.....
 [1]

- (b) State one similarity and one difference between gravitational field and electric field.

Similarity:

Difference:
 [2]

- (c) Fig. 2.1 shows the variation of the gravitational potential ϕ between the Moon and the Earth with distance from the centre of the Moon.

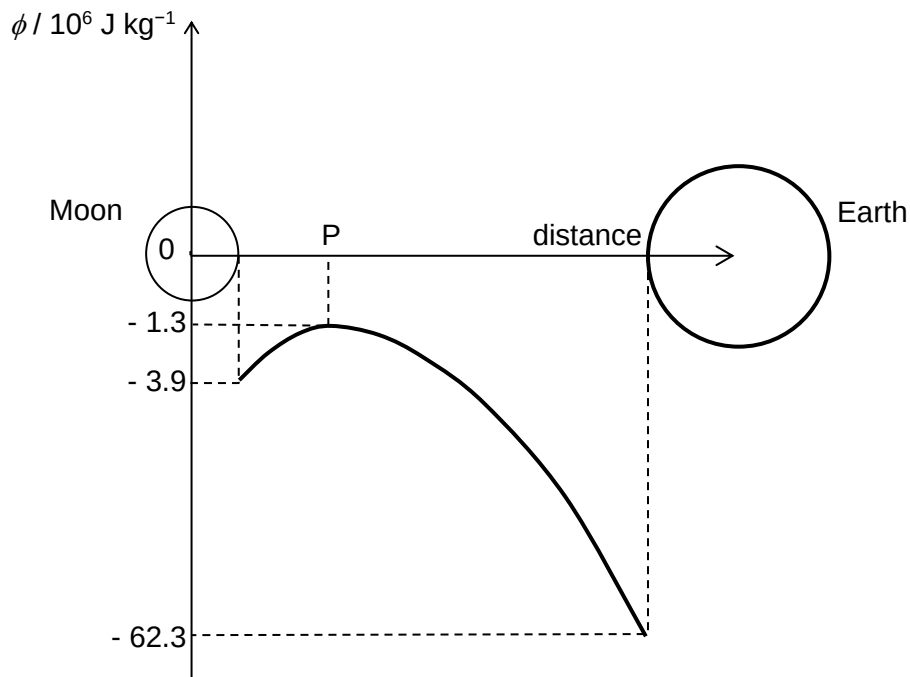


Fig. 2.1

- (i) Point P is a point along the line between the centres of the Moon and the Earth. Explain why point P is closer to the Moon than to the Earth.

.....

 [2]

- (ii) The distance between the centres of the Earth and the Moon is 3.80×10^8 m. Fig. 2.2 shows the values of the radius R and mass M of the Earth and Moon.

	R / m	M / kg
Earth	6.37×10^6	5.98×10^{24}
Moon	1.74×10^6	7.35×10^{22}

Fig. 2.2

Show that the distance from the centre of the Moon to point P is 3.79×10^7 m.

[2]

- (iii) NASA intends to project a space probe from the Earth to the Moon. Determine the minimum projection speed. Explain your answer.

speed = m s^{-1}

.....

[3]

- (d) Two small spherical charged particles P and Q are fixed in their positions at 5.0 cm apart in a vacuum. An electron is moved along the line joining the two charges as shown in Fig. 2.3.

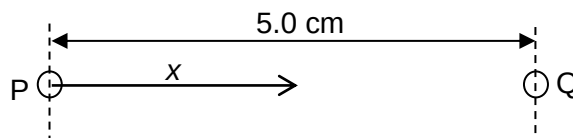


Fig. 2.3

The variation with the displacement x of electron from P of the electric potential energy E_p of the electron is shown in Fig. 2.4.

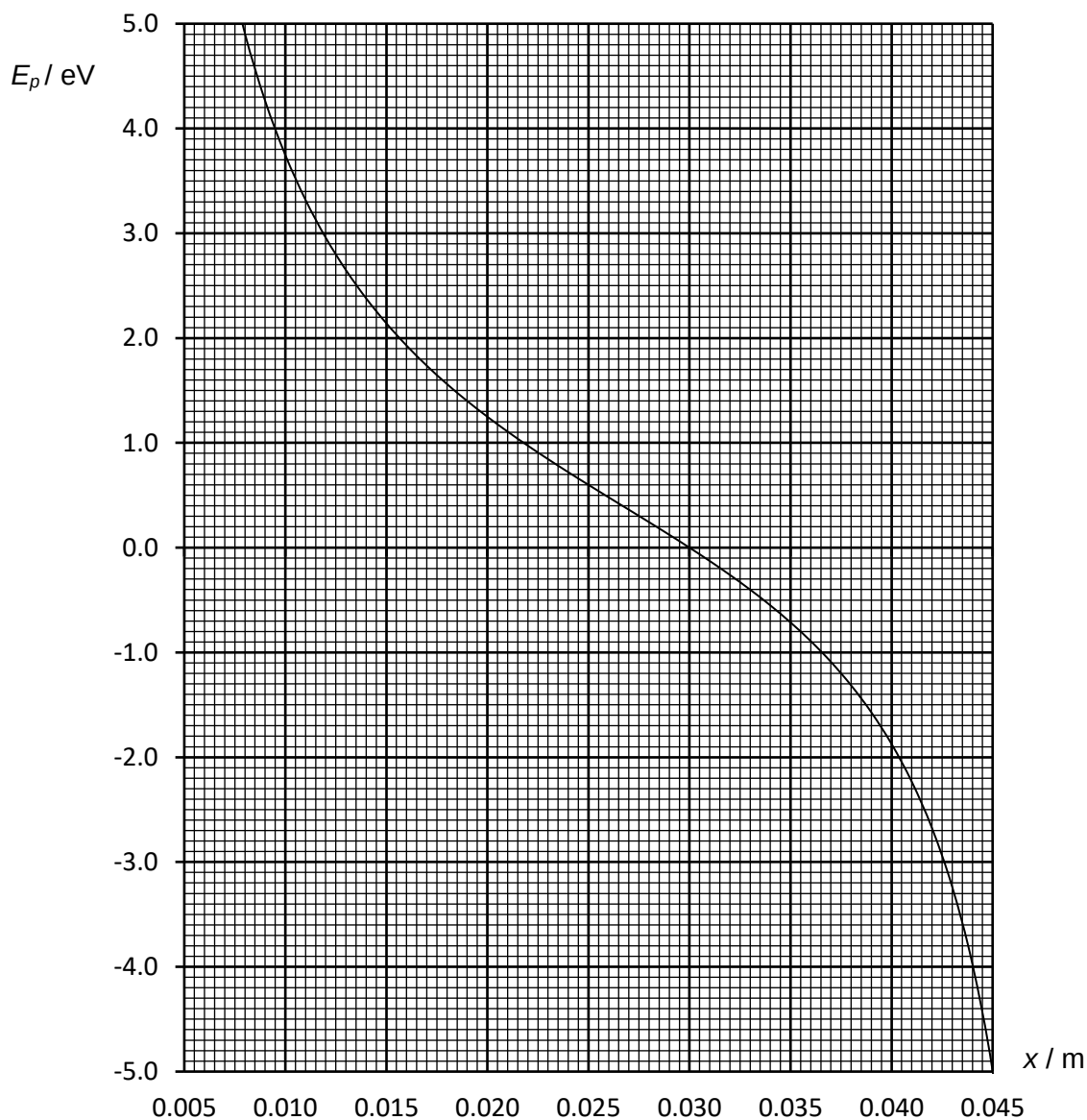


Fig. 2.4

- (i) State and explain which charge P or Q is negatively charged.

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.....

..... [2]

- (ii) Use Fig. 2.4 to estimate the magnitude of the force acting on the electron when it is at the point $x = 3.0$ cm.

force = N [3]

- (iii) On Fig. 2.4, sketch a new graph if the magnitude of charge Q is increased.

[1]